Project Proposal Prep

Mobile & Ubiquitous Computing 2026/2027

Ciências, Departamento de Informática

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Project

Themes

- **You propose**
- Has to be an app that only makes sense in a **fictional world** (e.g. Lord of the Rings)
  - ***Some of its features need to only make sense in that context***
- The app can be on any topic (e.g. health, navigation, shopping, games, sports, lifestyle, business, education etc)

Themes for Project

- **Every app as to be different**

Example (ChatGPT):

Quaffle Quest, the ultimate mobile app for witches and wizards to **track down missing** Quidditch balls! Using enchanted ball radar and **real-time alerts**, you'll never lose sight of the Quaffle, rogue Bludgers, or the elusive Golden Snitch—**on and off the pitch**. With advanced **features like Bludger detection**, **Snitch speed tracking**, and a m**agical Lost Ball Recovery Charm**, **Quaffle Quest**.

Whether you're playing, **coaching**, or just **spectating**, this app is your essential Quidditch companion

Project

Objective

Build an App or a set of Apps

Basic requirements

–Good user experience

–Feature rich (service integration, ..)

–Good data management

–Distributed

–Adequate combination of the above or others

Project Guidelines

- Build an application that could **only** be a **Mobile App**
- Leverage the **Device Sensors** to provide features that make sense for your app
- Handle gracefully **applications lifecycles** (i.e. stopping, resuming, starting)
- Leverage **multiple web API** to create a feature rich application
- **Multi-User** application (i.e. data and interactions shared across users)
- **Persistent Storage** (i.e. your app store and load relevant data)
- Create a **Dynamic** experience (e.g. events, push notifications, alerts)
- Integrate with mobile conventions (e.g. *share, material design)*
- **Be novel**! Do not build an app that already exists
- Make your app **accessible** (i.e. principles for accessibility)

Adequate combination of the above!

A Good Project Proposal

- It should motivate the concept
- Present related work (e.g. similar apps, papers)
- Describes the approach and features
    - Detail how some features are only present in the chosen fictional world
- Establishes a plan for the upcoming deadlines

See moodle examples of applications built in previous years

Alternative Project

- A Mobile application for the FCUL's Dia Aberto / Open Day
- Project basic requirements and guidelines are the same
- May have a few required features, but room for students' creativity
- Max of 3 Groups on the alternative project (selection based on the proposal)
  - Talk with me if planning to propose such project
- Potential to continue after the CM Course to make it "official"

Move Together: Community-Based Physical Activity in Torres Vedras

How can mobile technology help a city become more active together?

This project will develop a mobile app and supporting platform that **enable residents of Torres Vedras to track physical activity, join neighbourhood challenges, and work towards shared community goals**. Activities could include collaborative walking targets, team competitions, and routes connecting local points of interest. Students will explore smartphone sensors, location services, activity visualisation, and contextual notifications to support participation. A central challenge is designing an inclusive experience that accommodates different abilities and fitness levels, gives people control over what they share, and encourages sustained engagement through community connections.

ADR-Prevent: Making Personalised Medication Information Understandable

How can a mobile app help people understand their pharmacogenomic profile and participate in monitoring medication safety?

This project will develop a prototype that **explains how an individual's genetic profile may affect their response to selected medications, presenting information about potential adverse drug reactions (ADRs) in accessible language**. Users will be able to maintain a medication list, scan medication packaging, receive contextual prompts, and report suspected reactions. A complementary interface will allow physicians and pharmacists to consult relevant information and, with consent, access individual profiles. Students will explore personalised interfaces, understandable risk communication, consent management, and mobile reporting workflows, using expert-approved content.

Promotion and Monitoring of Physical Activity

How can a mobile app leverage smartphone sensors and community resources to promote physical activity and monitor progress in people at risk of type 2 diabetes?

This project proposes the development of a mobile application to **promote physical activity, particularly among people at risk of developing type 2 diabetes**. The application should allow users to browse exercise programs and resources available in their community, set personal physical activity goals, and track their progress over time. Goal is to explore the use of smartphone sensors to automatically monitor **physical activity indicators, such as step count and periods of activity or inactivity.** The application may also use the user's location to recommend nearby exercise resources and provide contextualized notifications or suggestions that encourage regular physical activity. Features that explore how to leverage the community could also be interesting.. The work should result in a functional prototype that integrates monitoring mechanisms, data visualization, and contextual interaction. The objective is not to develop the complete clinical ecosystem, but to explore and implement some of its most relevant mobile features from the user's perspective.

In three weeks (04/11/2026) - Pitch

All group members should be in the same TP! Talk with me if there are issues or questions.

Presence in mandatory

A (**5 minute max**) presentation that you will then present in class.

It should have, at least:

- Title and group identification
- Motivation
- Concept and background (related work)
- Main envisioned features
- Technologies (which sensors, APIs, algorithms, etc... what are you planning to include)
- Planning

Optional

- Preliminary design (Low-fidelity prototypes)
- Evaluation (plan on how to evaluate)

Submit the presentation on the day before (03/11/2026)